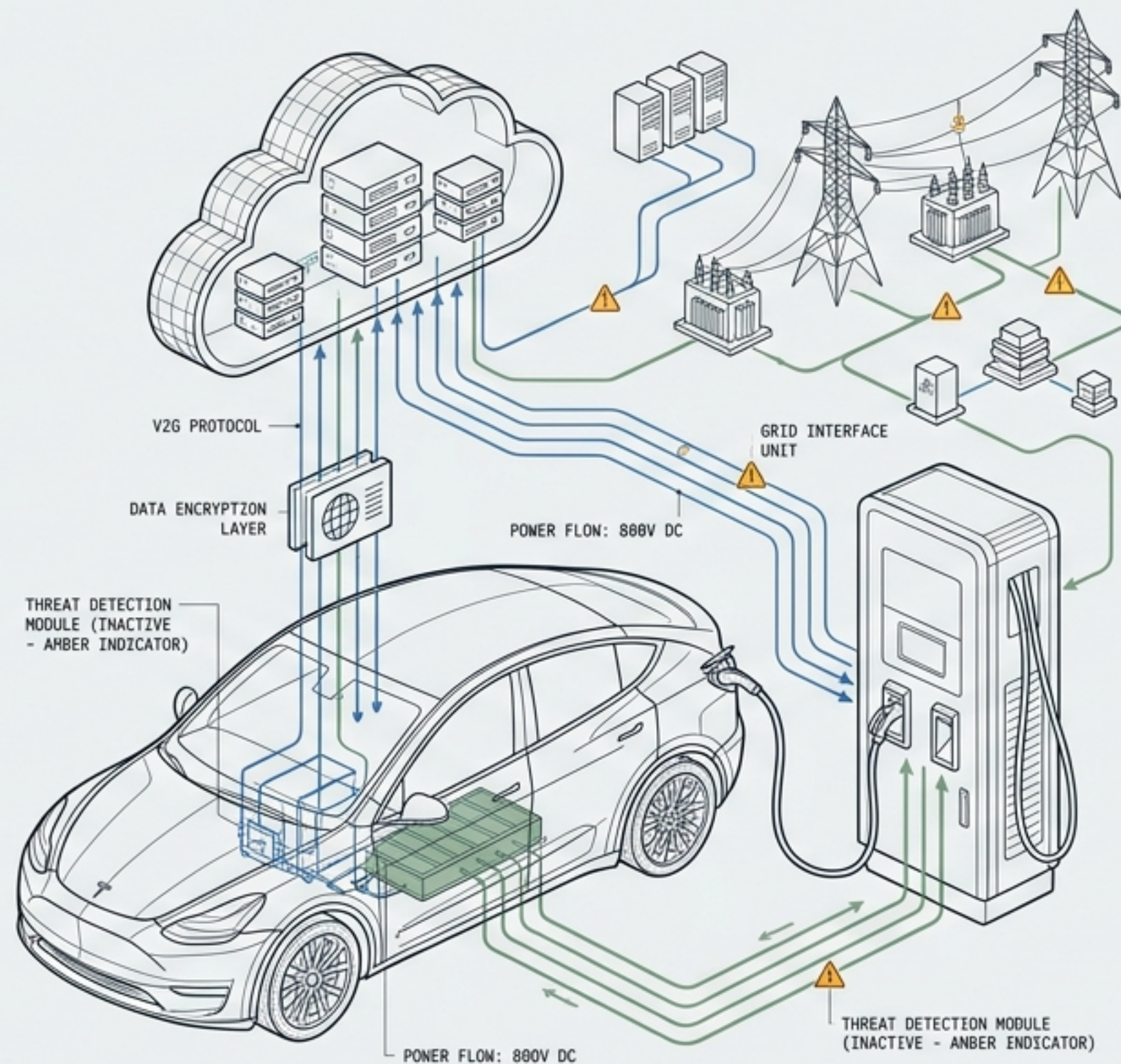


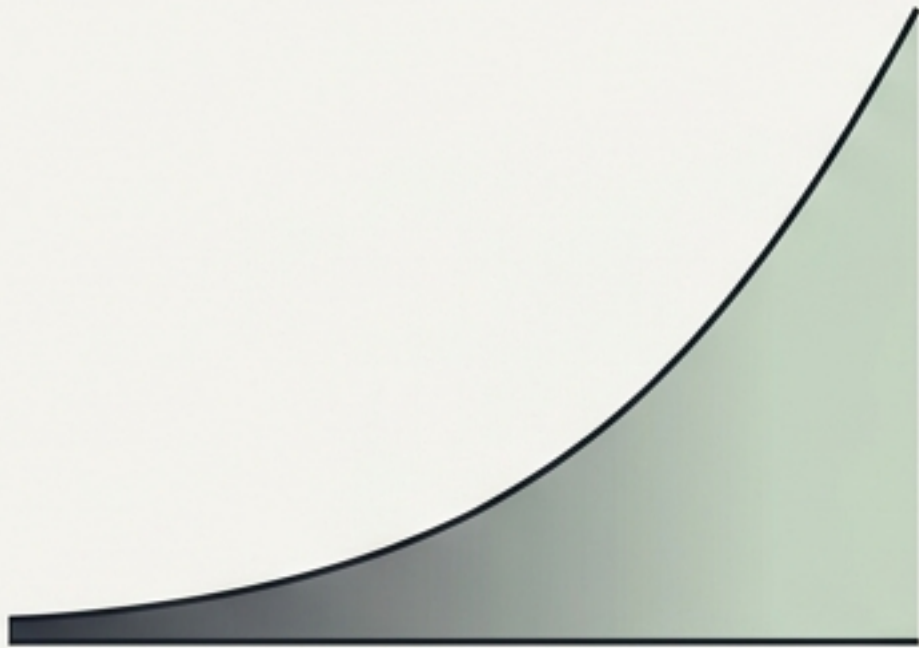
Securing the Grid's Edge: The Future of EV Charging Cybersecurity

Moving from Siloed Defenses to Adaptive, Cyber-Physical Immunity



The High-Stakes Convergence

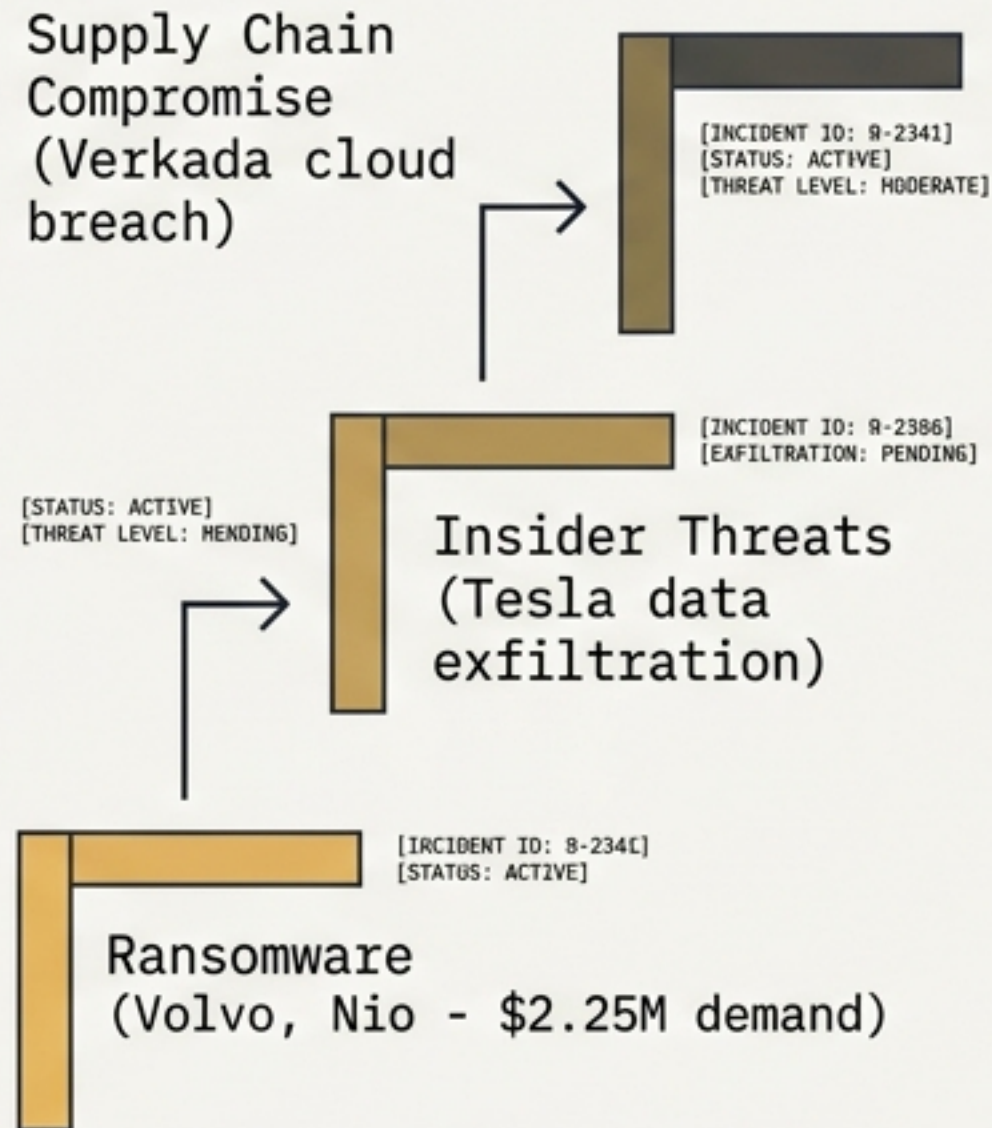
Market Velocity



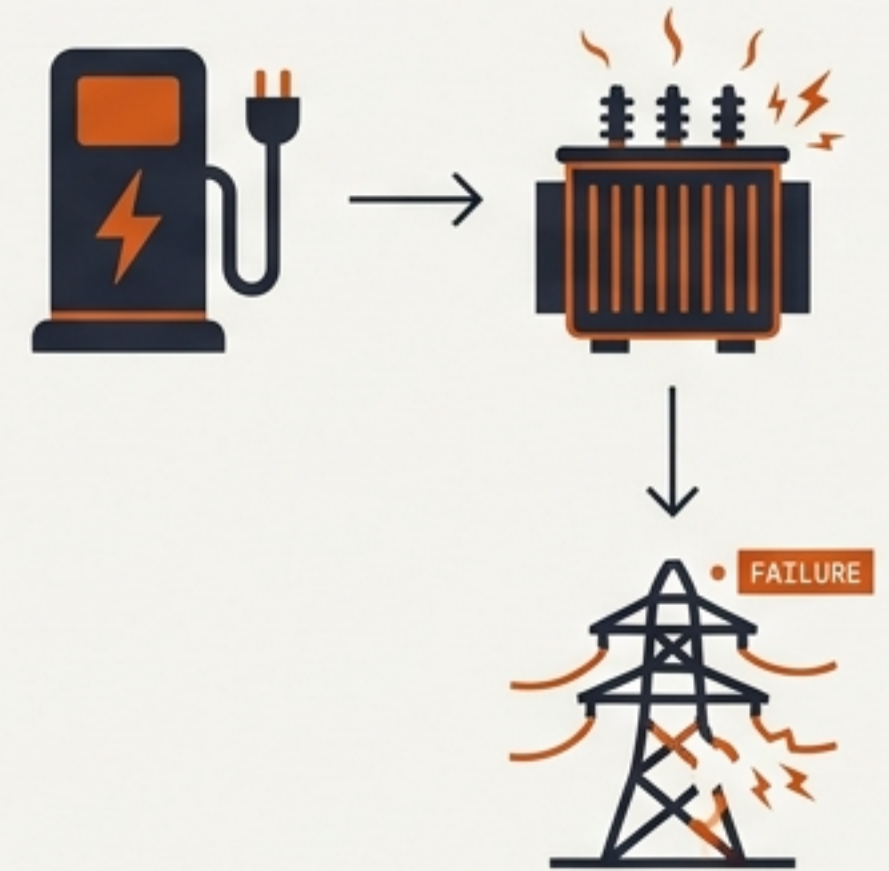
Projection: 16 Million
EV sales/year by 2027

Market Capitalization:
\$457B (2023)

The Evolving Threat

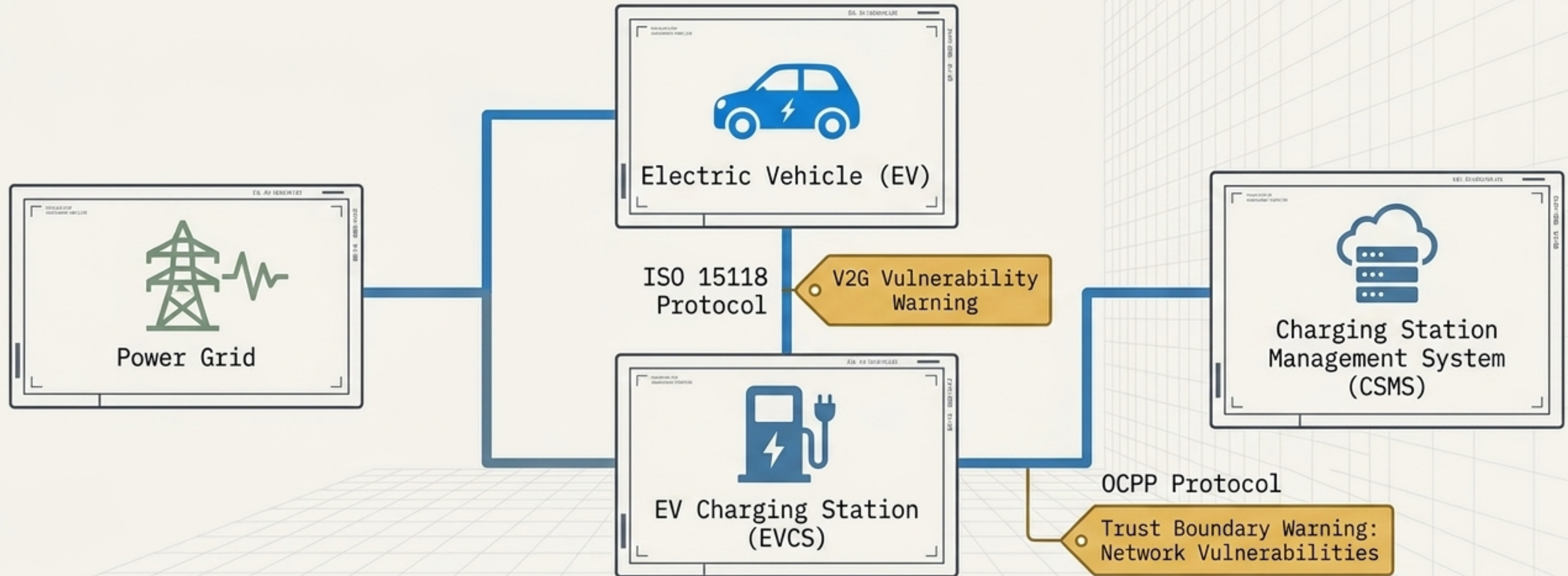


The Ultimate Risk



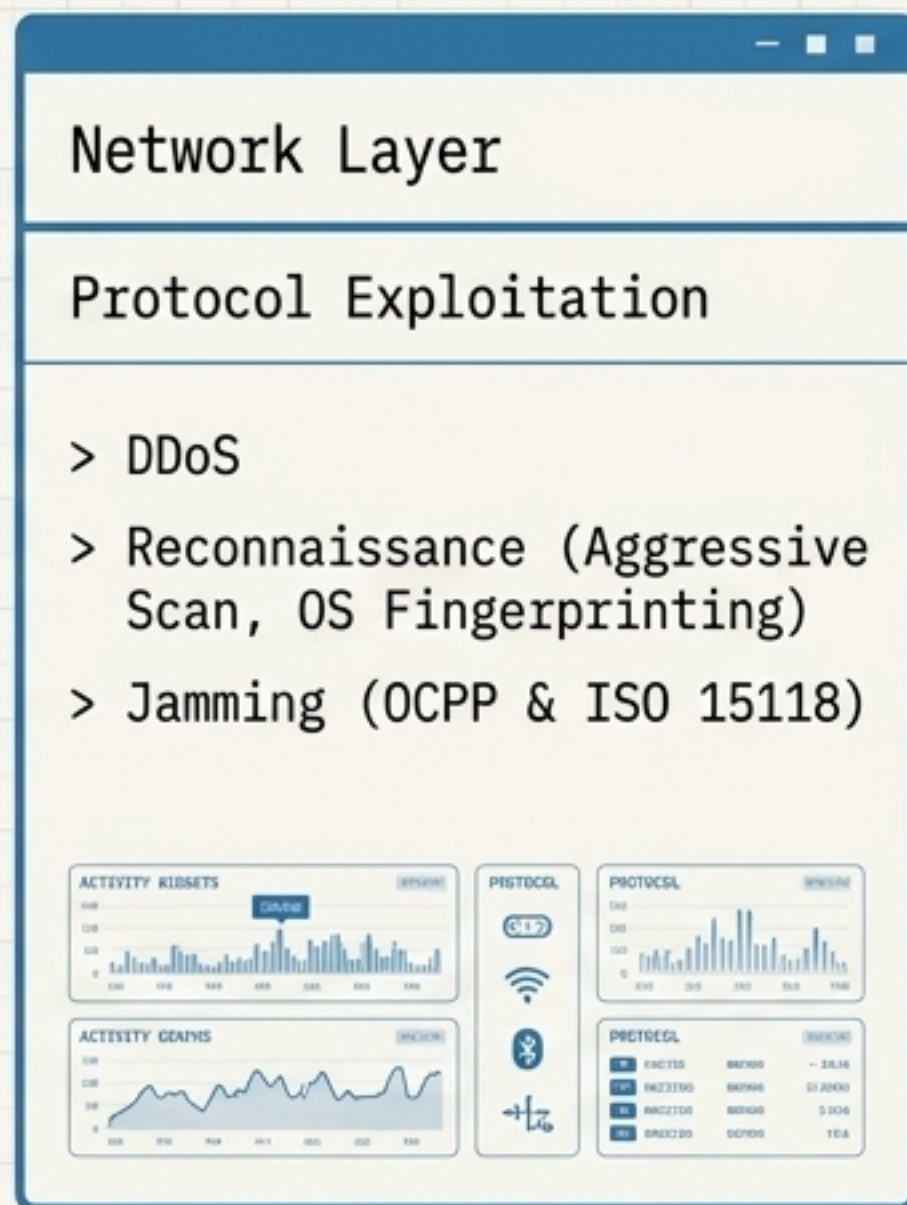
Compromising the charging network edge can trigger [VOLTAGE FLUCTUATIONS], [TRANSFORMER OVERLOAD], and [NATIONAL POWER GRID DESTABILIZATION].

The EV Cyber-Physical Ecosystem

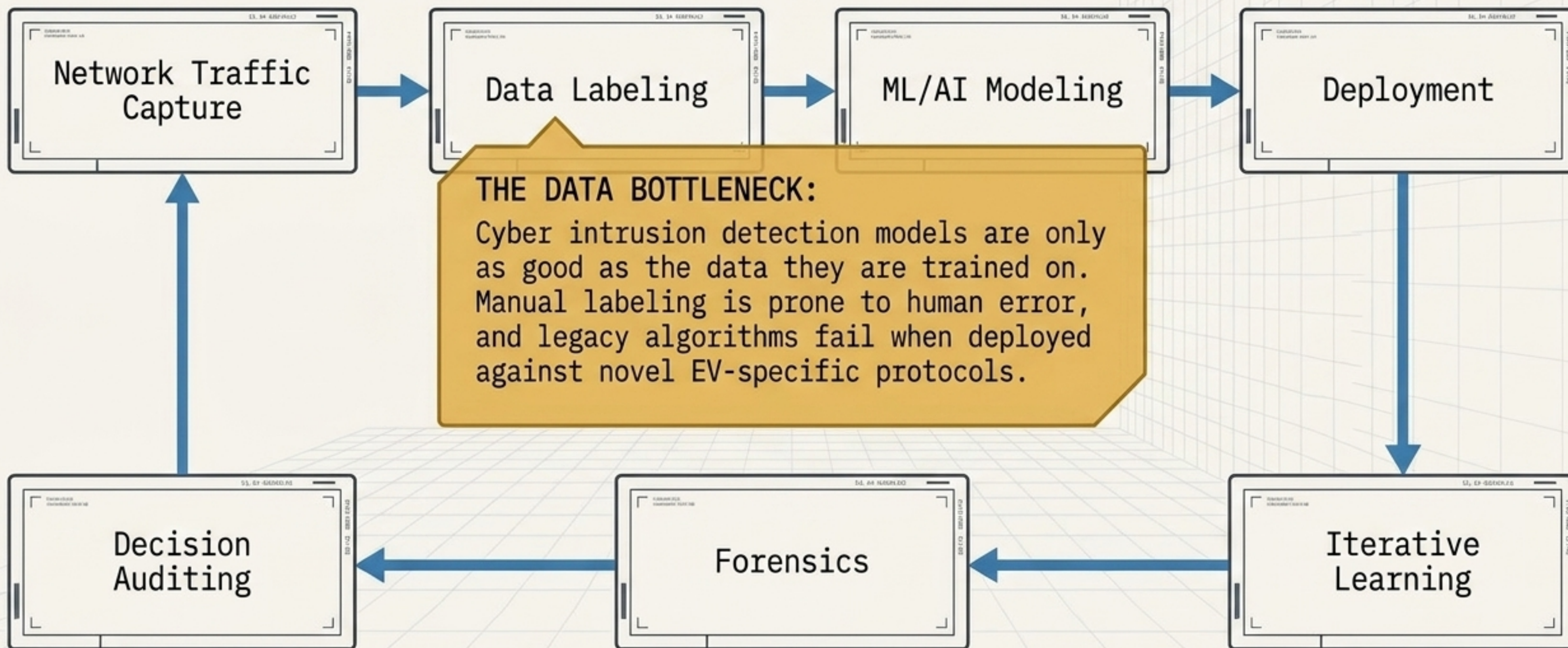


Diagnostic: The EV ecosystem is a cyber-physical system where digital communications directly control physical power flows. Every protocol bridge is a potential attack surface.

Anatomy of EV Cyberattacks



The First Line of Defense: Intrusion Detection Systems (IDS)



The Training Data Dilemma

Before (Legacy Approach)

Data Sources:

Generic IoT datasets (e.g., IoT-23) or completely simulated network attacks.

Blindspots:

Three 2 EV network ampresates

- ✗ Lacks application-layer visibility
- ✗ Missing benign EV charging session data
- ✗ Entirely disconnected from physical power flows

Now (The CICEVSE2024 Standard)

Data Sources:

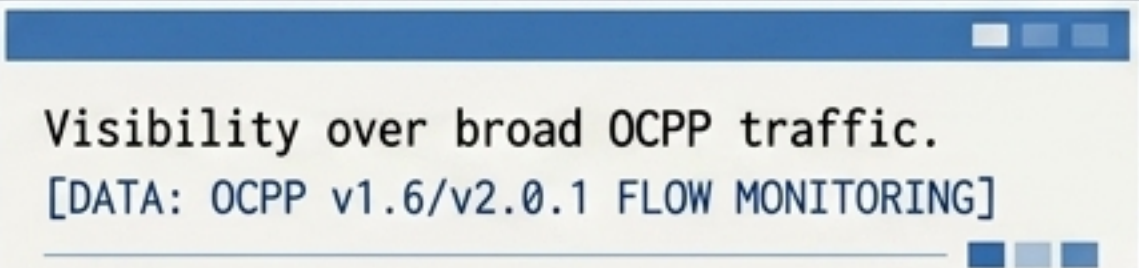
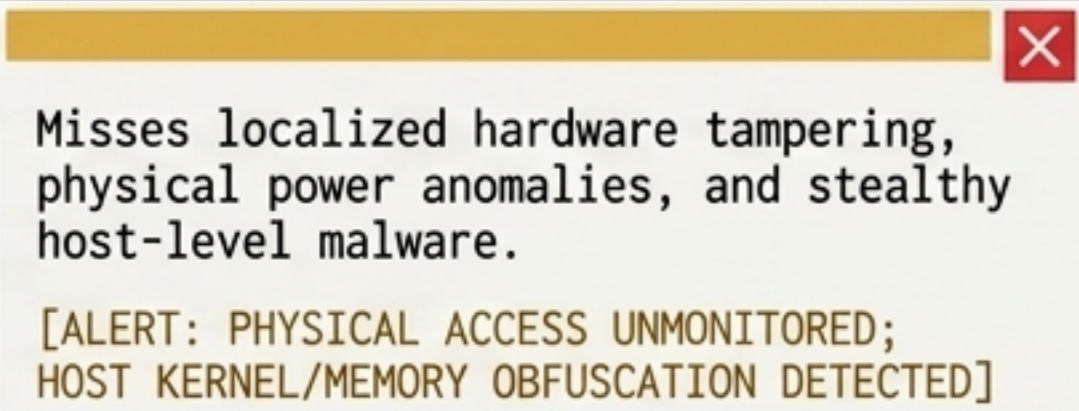
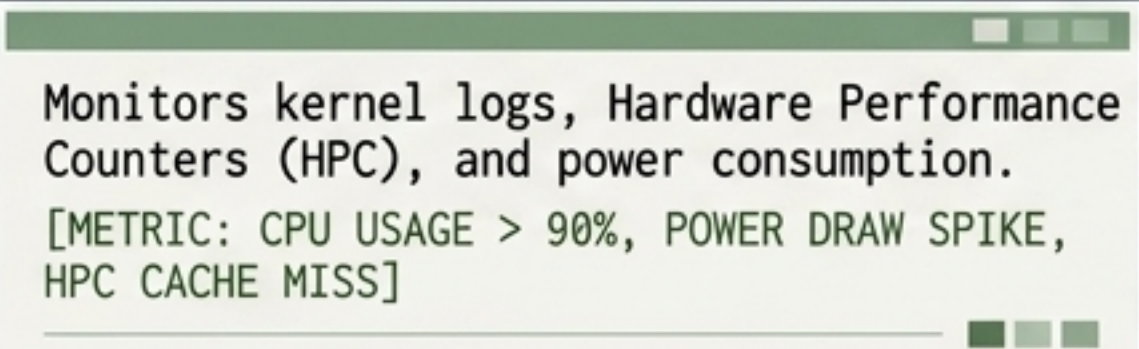
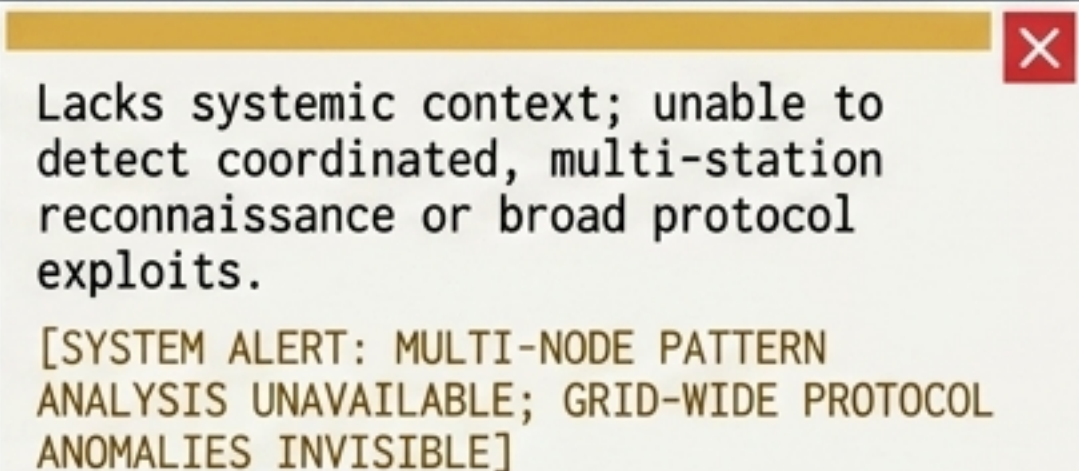
Multi-dimensional capture using real hardware (EVSE-A) and Raspberry Pi emulators (EVSE-B).

Capabilities:

- ✓ Flow-level network traffic
- ✓ Packet-level payloads
- ✓ Packet-level payloads
- ✓ Kernel-level Host Events
- ✓ Physical Power Consumption metrics

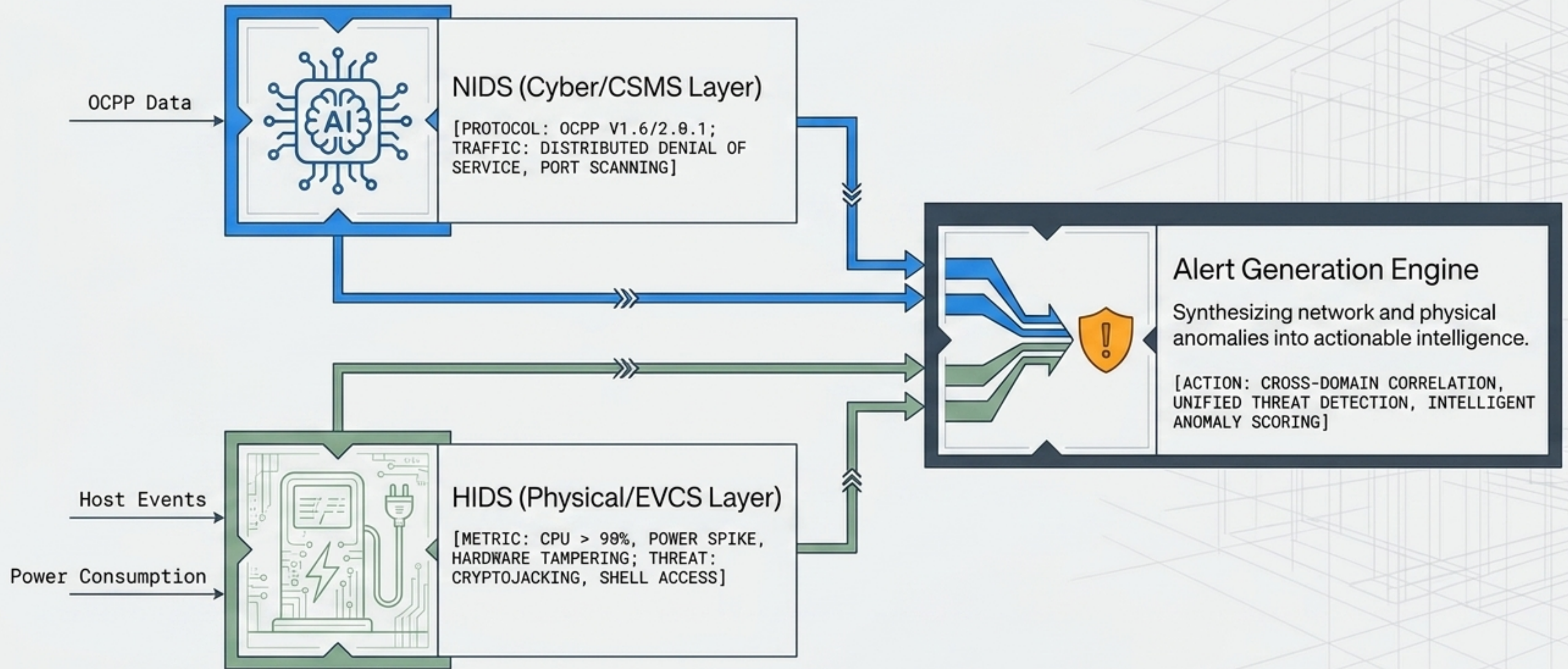
Takeaway: True cyber-physical anomaly detection requires training on integrated network, host, and electrical data.

The Defense Gap: Network vs. Host

NIDS (Network-Based IDS) Deployment: At the CSMS cloud level.	 Visibility over broad OCPP traffic. [DATA: OCPP v1.6/v2.0.1 FLOW MONITORING] Excellent at catching distributed DoS and network reconnaissance. [PROTOCOL: TCP/IP SYN FLOOD, PORT SCANNING]	 Misses localized hardware tampering, physical power anomalies, and stealthy host-level malware. [ALERT: PHYSICAL ACCESS UNMONITORED; HOST KERNEL/MEMORY OBFUSCATION DETECTED]
HIDS (Host-Based IDS) Deployment: At the individual EVCS edge.	 Monitors kernel logs, Hardware Performance Counters (HPC), and power consumption. [METRIC: CPU USAGE > 90%, POWER DRAW SPIKE, HPC CACHE MISS] Excellent for Cryptojacking and Backdoors. [THREAT: MINING SCRIPT EXECUTION, UNAUTHORIZED SHELL ACCESS]	 Lacks systemic context; unable to detect coordinated, multi-station reconnaissance or broad protocol exploits. [SYSTEM ALERT: MULTI-NODE PATTERN ANALYSIS UNAVAILABLE; GRID-WIDE PROTOCOL ANOMALIES INVISIBLE]

Takeaway: A siloed approach leaves the ecosystem fundamentally exposed.
[CRITICAL INSIGHT: UNIFIED DEFENSE ARCHITECTURE REQUIRED FOR COMPLETE VISIBILITY]

Breakthrough 1: The Hybrid IDS Architecture



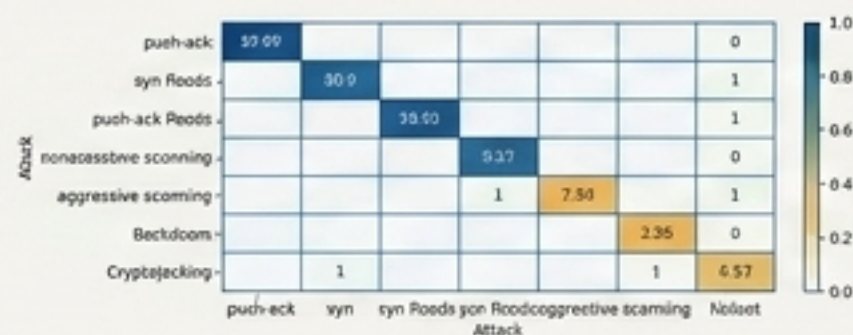
Hybrid IDS Performance Diagnostics

NIDS Telemetry (Network)

99.99% Accuracy

(Powered by XGBoost)

Near-perfect detection of push-ack floods, syn floods, and aggressive scanning. Only 5 misclassifications out of 54,784 samples.



HIDS Telemetry (Host/Power)

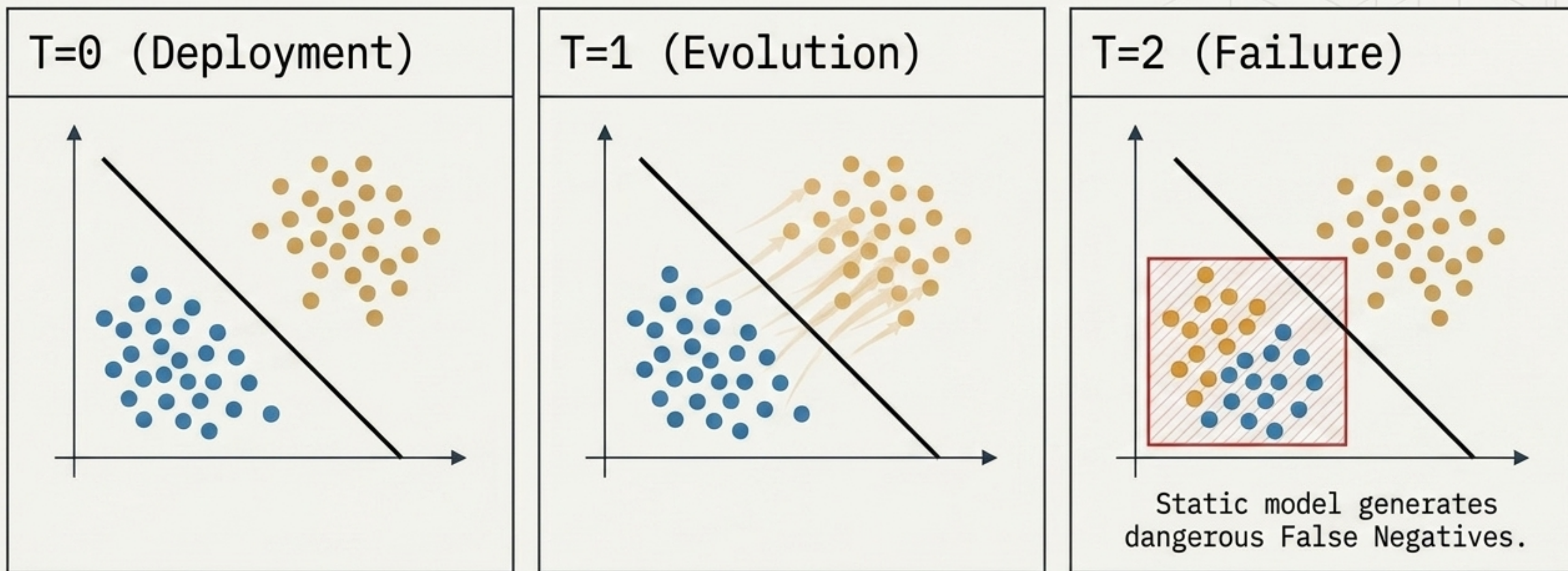
83.47% Accuracy

F1-Scores:

- 0.96 False Data Injection
- 0.96 Backdoors
- 0.97 Cryptojacking

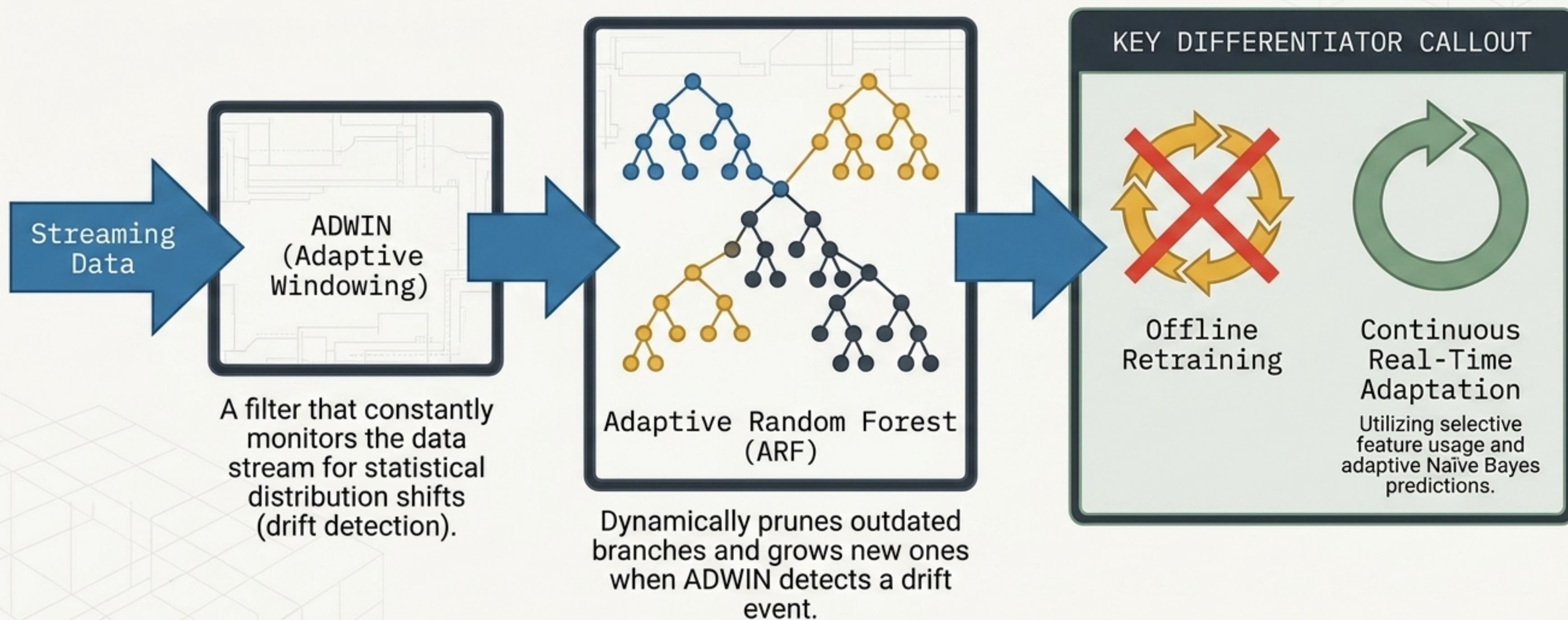
Dual-layer detection successfully correlates network-level distributed attacks with physical, host-level consequences.

The Hidden Threat of Concept Drift

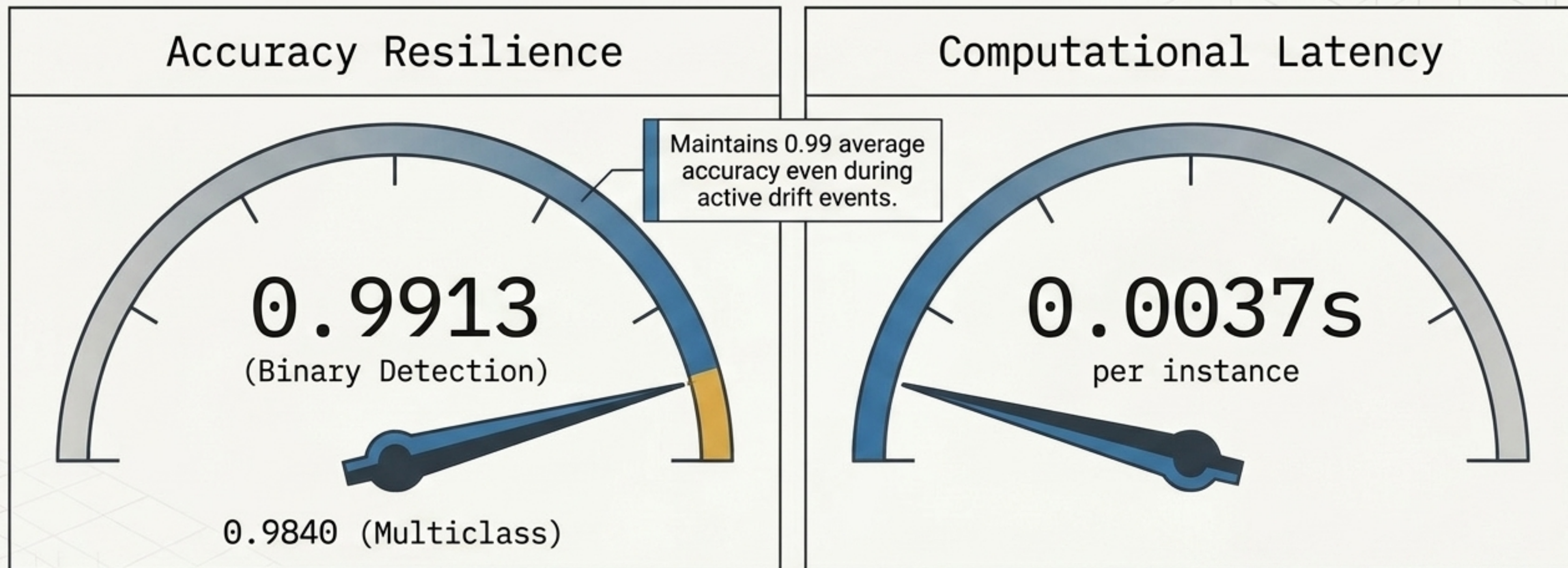


The Hidden Threat: Cybersecurity is an arms race. Static offline models degrade rapidly as adversaries adapt their protocols and behaviors.

Breakthrough 2: Online Machine Learning



Real-Time Viability: Speed & Accuracy



Real-Time Edge Viability: Online Machine Learning delivers high-fidelity threat detection with a computational footprint small enough to deploy directly on resource-constrained EVCS hardware.

Beyond the Machine: Insider Threats & Supply Chain

Privilege Escalation & Ghost Accounts



Case: Martin Tripp (Tesla)

Unmanaged local admin rights and dormant accounts act as exfiltration vectors. Endpoint detection systems routinely fail against legitimate administrative credentials.



Third-Party Supply Chain



Case: Verkada Cloud Breach

Unvetted vendor super-admin rights can compromise entire fleets of connected devices.

Security must evaluate the business case, not just the technical integration.



Data Exfiltration via Insiders

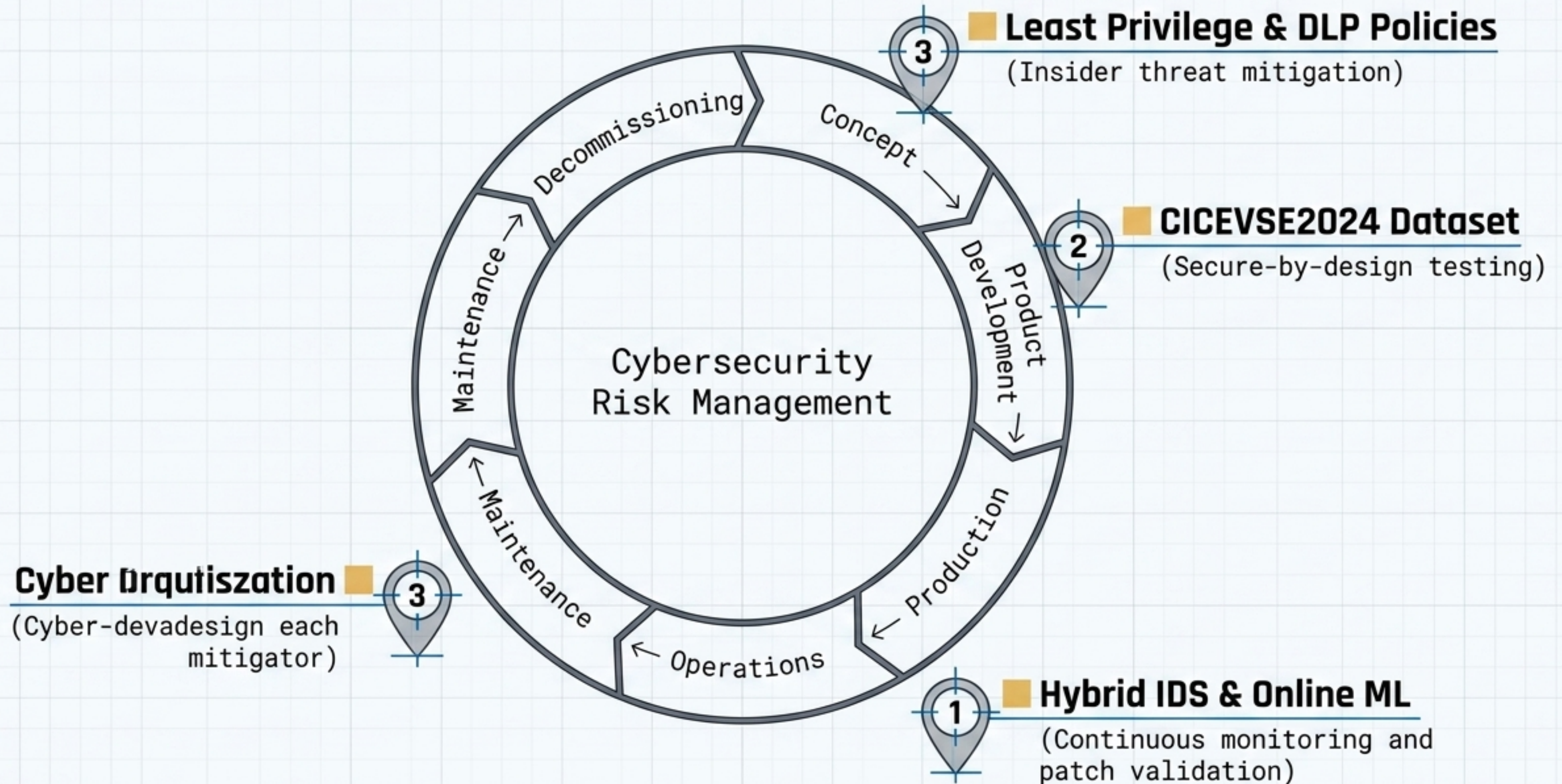


Case: Alex Khatilov (Tesla)

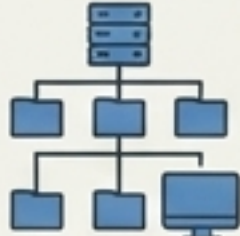
Standard background checks miss dark web activity. Mitigation requires strict Data Loss Prevention (DLP) and the enforcement of the Principle of Least Privilege.



Governance & Cyber Resilience (ISO/SAE 21434)



The Cyber-Physical Blueprint

Architecture (The Where) 	Intelligence (The How) 	Governance (The Who) 
Dual-Layer Hybrid IDS 	Online Machine Learning 	ISO 21434 & Zero Trust 
Unifying Network (CSMS) and Host (EVSE) visibility to eliminate blindspots and bridge the cyber-physical gap.	Utilizing Adaptive Random Forest (ARF) and ADWIN to combat concept drift and continuously learn at the edge in real-time.	Enforcing the principle of least privilege, strict vendor risk assessments, and continuous auditing across the supply chain.

True security at the grid's edge requires systems that learn exactly as fast as the threats evolve.